

Direction recovery



Pearson- δ
 δ -cosine

does the prediction
point the right way?

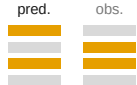
Allele discrimination



$d(\text{own}) < d(\text{others})$

does it identify its
own allele?

DE fidelity



DE overlap
DE-LFC rank corr.
direction agreement

do the changed
genes agree?

Central question: can models distinguish alleles, not merely
recover the shared direction of a perturbed gene?